Weighted Graph-Based Event Clustering and Priority Scoring for Flood-Rescue Coordination Using Edge AI

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Abstract. During flood disasters, telecommunication infrastructure often collapses, disabling cloud-centric processing exactly during the rescue "golden hour." We propose an end-to-end framework combining Edge AI and weighted graph theory to collect, cluster, and automatically rank rescue events. Edge devices extract a multidimensional attribute vector (L, T, F, E, N, V, C) from images and text, then transmit only a sub-kilobyte metadata packet instead of raw media. On the server, events are modeled as a weighted graph in which geographic distance acts as a *multiplicative gate* rather than an additive term, guaranteeing that every cluster is spatially coherent. The Louvain algorithm (with Leiden recommended) decomposes the graph into "operational zones," and a cluster-level priority function $\mathcal{P}(C_k)$—built from a normalized risk core and a demographic-vulnerability factor acting as an *amplifier*—ranks clusters to support dispatch. On a deterministic synthetic dataset of 285 events modeled on Central Vietnam, the gating form reduces mean cluster diameter from **100 km to 0.30 km** while preserving clustering accuracy (Adjusted Rand Index, ARI = 0.892); a confidence gate C_i suppresses fake reports that inflate victim counts (a **55%** reduction of phantom population); on the same gating graph the framework attains ARI 0.892 versus Spectral Clustering (0.339) and HDBSCAN (0.890 but 25 km diameter), and beats geometric baselines K-Means (0.688) and DBSCAN (0.730) on raw coordinates; priority rankings are stable (Kendall's τ: mean 0.99, min 0.94 under ± 0.10 weight perturbation).

Keywords: Edge AI · Event clustering · Weighted graph · Community detection · Louvain · Rescue prioritization · Multimodal analysis · Flood disaster.

1 Introduction

Climate change is intensifying the frequency and severity of extreme weather events. Vietnam—with its long coastline and terrain directly exposed to typhoon circulation—is affected by roughly 6–8 typhoons and tropical cyclones each year (out of the ∼11 storms and tropical depressions that develop in the East Sea) [11], causing heavy losses of life and property, especially in the central and northern regions.

A core cause of failure in emergency response is the disruption of telecommunication infrastructure. When the power grid degrades and base transceiver stations (BTS) become isolated, cloud-centric data collection and processing pipelines fail entirely, cutting command centers off from the storm's epicenter precisely during the "golden hour." In this context, social media and messaging applications become a survival-critical *social sensing* channel, producing a *multimodal* data stream (text, images, video, spatiotemporal metadata) that is fragmented, duplicated, noisy, and prone to *information overload* [10].

This paper proposes an end-to-end framework that simultaneously addresses three challenges:

1. **Surviving degraded networks:** push AI to edge devices; instead of uploading megabytes of images/video, the application processes on-device and transmits only a sub-kilobyte metadata packet.
2. **Physically meaningful de-duplication of events:** build a weighted graph integrating space, time, and hazard context, then perform community detection.
3. **Automatic priority ranking:** quantify urgency at the *cluster* level to answer the dispatch question "which cluster first?"

Main contributions.

- A **multiplicative (gating) edge-weight function** in which geographic similarity globally modulates non-spatial similarity, guaranteeing spatially coherent clusters—a prerequisite for dispatching boats with a finite operating radius.
- A **cluster-level priority function** $\mathcal{P}(C_k)$ with a normalized risk core and a **demographic-vulnerability factor as a multiplicative amplifier**, injecting equity into the resource-allocation problem.
- Two **edge-feasible** auxiliary attributes: a vulnerability index V_i (folded into the existing text classifier) and a confidence score C_i (a lightweight sigmoid heuristic), both providing robustness against fake reports.
- A comprehensive **empirical validation** on a deterministic synthetic dataset modeled on Central Vietnam, justifying each design decision with quantitative evidence.

2 Related Work

2.1 Multimodal Analysis in Crisis

The shift from unimodal to multimodal learning has substantially improved the situational awareness of disaster-management systems. Pioneering datasets such as CrisisMMD [1] and FloodNet [17] provide training foundations for damage assessment and urgency classification. For images/video, convolutional networks such as ResNet and MobileNetV3 [8], or semantic-segmentation models, are used to extract flooded regions; some work applies human pose estimation to infer water depth where no physical gauges exist. For text/audio, Transformer architectures (BERT, DistilBERT [20]) perform sentiment analysis, entity recognition, and needs identification. Graph-neural-network approaches such as CrisisSpot [4] show that fusing modalities improves disaster-content classification F1 by 5.01–9.45%.

2.2 Edge Computing for Disaster Response

The critical bottleneck of multimodal models is their bandwidth and compute demand. During floods, uploading high-resolution media to the cloud is infeasible. The community therefore promotes *Edge AI* [13,15]: model compression (quantization, knowledge distillation) and lightweight architectures enable on-device inference. The edge device transmits only a compact metadata packet (sub-kilobyte, quantified in Sect. 6), so distress signals still penetrate congested networks. EmergencyNet [13] exemplifies a lightweight CNN for drone/embedded platforms with millisecond-scale inference latency.

2.3 Graph-Based Spatiotemporal Analytics

Treating each distress call as an isolated point discards global context. Graph theory offers a mathematical framework to model interactions and risk propagation: a rescue event is a vertex v in a graph $G = (V, E, W)$, with edges E encoding relationships and weights W their strength. Event-detection studies typically build weighted graphs from spatiotemporal proximity and keyword co-occurrence (e.g., TF-IDF similarity). Advanced models incorporate Euclidean/Haversine distance to *penalize* links between distant events, forming geo-semantic graphs—yet they still omit flood depth and demographic vulnerability.

2.4 Community Detection

Traditional unsupervised methods are limited: **K-Means** [14] requires a predefined cluster count K (infeasible in a volatile disaster) and assumes spherical clusters; **DBSCAN** [5] is sensitive to high dimensionality and parameters. Network-topology methods, notably **Louvain** [2], are the gold standard for weighted-graph clustering via optimization of the **Modularity** function [16], with observed near-$\mathcal{O}(N \log N)$ runtime. The **Leiden** [21] variant fixes the badly-connected-community pathology that Louvain occasionally exhibits. A broader survey of the field is given by Fortunato [6].

2.5 Positioning

Table 1 positions this work against representative studies on five capability axes. No prior work simultaneously (i) runs multimodal extraction at the edge, (ii)

Table 1. Positioning against representative work (✓ yes, ∼ partial, blank no).

Study	Multimodal	Edge/on-device	Weighted graph	Vulnerability	Cluster priority
CrisisSpot [4]	✓		✓ (GNN)		
Event detection (TF-IDF)			✓		
EmergencyNet [13]		✓			
Vulnerability prioritization [7]				✓	∼
Proposed framework	✓	✓	✓ (gating)	✓ (factor)	✓

encodes it into a spatial–semantic–physical gating weighted graph, (iii) integrates demographic vulnerability as an amplification factor, and (iv) produces a cluster-level priority ranking. That empty cell is the gap this paper fills.

3 Research Gaps

A literature review reveals three fundamental gaps, motivating the proposed framework.

Gap 1 — Missing hazard-specific attributes in edge weighting. Two geographically close events do not necessarily share the same risk, which is also governed by micro-topography, building structure, and victim condition. Omitting flood depth (from images) and urgency (from text) from the edge weight prevents the graph from distinguishing a group trapped on a rooftop from a group safe on a high floor at the same coordinates.

Gap 2 — Vulnerability blind spot. Humanitarian-logistics frameworks often assume homogeneous demand and minimize only travel distance/time [22]. But disasters impact groups unequally: the elderly, children, pregnant women, and the disabled deteriorate faster. Failing to *amplify* priority for events containing vulnerable individuals is a serious ethical (equity) shortcoming [7].

Gap 3 — Static clustering, no community-level priority. Many studies stop at "detecting event groups." But coordinators need to know which cluster to dispatch to *first*. A cluster-level priority mechanism—combining maximum flood depth, total trapped count, vulnerable fraction, and overall urgency—has not been rigorously formalized.

4 Proposed Framework

The framework has four sequential stages: (4.1) edge extraction of a multidimensional attribute vector; (4.2) construction of a spatial–semantic–physical weighted graph; (4.3) community decomposition via Louvain/Leiden; (4.4) cluster-level priority scoring.

4.1 Multidimensional Attribute Vector

Each rescue event v_i is represented by seven attributes:

$$v_i = (L_i,\ T_i,\ F_i,\ E_i,\ N_i,\ V_i,\ C_i) \tag{1}$$

Table 2. Proposed vulnerability label weights and their mapping to the dataset's four discrete V_i levels.

Detected label(s)	V_i level	Rationale
None	0.0	No vulnerable individual reported
Elderly *or* disabled (one)	1.0	Single standard vulnerable individual
Pregnant woman, or two labels	1.5	Elevated medical urgency / compounding
Infant, or three+ labels	2.0	Highest deterioration rate

where L_i is GPS location, T_i a timestamp, $F_i \in [0, 1]$ physical flood level (MobileNetV3 segmentation/pose estimation), $E_i \in [0, 1]$ urgency (DistilBERT/UIT-VSMEC text sentiment), $N_i \in \mathbb{Z}^+$ the trapped count, $V_i \geq 0$ a demographic-vulnerability index, and $C_i \in (0, 1)$ an information-confidence score. The edge device transmits only a JSON string of these values (sub-kilobyte; Sect. 6) instead of megabytes of raw media.

Edge feasibility. The base attributes L, T, E, N, F follow the project's commitments (images via MobileNetV3, quantized DistilBERT for text). The two auxiliary attributes are designed to add *no* heavy deep model: V_i is extracted by a multi-label head sharing the same text classifier used for E_i (detecting phrases such as "infant present," "exhausted elderly," "pregnant woman"), and C_i uses a lightweight sigmoid heuristic. Full versions (dedicated crowd counting/pose estimation for N_i; a confidence model learned from user history for C_i) are explicitly positioned as future work beyond the six-month prototype.

Operational definition of V_i. V_i is the sum of humanitarian-priority weights over the vulnerable labels detected in a report. Table 2 gives the *proposed* weight scheme and its mapping to the four discrete levels $\{0, 1, 1.5, 2\}$ used in the dataset. We stress a scope boundary: the experiments assume V_i is *given* (ground-truth vulnerability), so they test the amplification *mechanism* of $\mathcal{V}_{agg}$, *not* the capability to extract V_i from raw text—the latter is future work. The weight table is a normative proposal (who sets it is a policy decision for command staff), not an empirically optimized artifact.

Confidence heuristic.

$$C_i = \sigma\big(b_0 + b_1 \cdot \mathbb{1}[\text{has image}] + b_2 \cdot \log(1 + n_i^{\text{corrob}})\big) \quad (2)$$

Here σ is the sigmoid, $\mathbb{1}[\text{has image}]$ indicates vision-verified multimodal evidence, and n_i^{corrob} counts independent corroborating reports within a corroboration radius $r_{\text{corrob}} = 400\,\text{m}$ and time window $\Delta t_{\text{corrob}} = 60\,\text{min}$ (kept deliberately separate from the clustering constants $\sigma_{geo}, \tau_{temp}$ to avoid cross-coupling). Log compression makes the 2nd–3rd corroborating report boost confidence strongly while the 50th adds almost nothing, resisting single-location spam. In experiments we use $(b_0, b_1, b_2) = (-0.2,\ 1.4,\ 0.9)$. Absent a persistent-account infrastructure, "independence" is approximated purely by spatiotemporal proximity;

we return to the resulting vulnerability to coordinated fake corroboration in the adversarial analysis of Sect. 5.9.

4.2 Spatial–Semantic–Physical Weighted Graph

This is the most important design fix relative to a naive draft. A *naive additive* weighting

$$w_{ij} = \alpha \mathcal{S}_{geo} + \beta \mathcal{S}_{temp} + \gamma \mathcal{S}_{context} \tag{3}$$

fails: two events 50 km apart ($\mathcal{S}_{geo} \approx 0$) but both describing "water up to the roof" ($\mathcal{S}_{context} \approx 1$) still receive a substantial $w_{ij} \approx \gamma$. The clustering algorithm would then merge distant points into one "operational zone"—meaningless for a boat with a finite radius.

We instead adopt a **multiplicative (gating)** form in which $\mathcal{S}_{geo}$ sits *outside* as a global modulating factor:

$$w_{ij} = \mathcal{S}_{geo}(L_i, L_j) \cdot \Big(\beta \cdot \mathcal{S}_{temp}(T_i, T_j) + \gamma \cdot \mathcal{S}_{context}(v_i, v_j)\Big) \tag{4}$$

When distance is large, $\mathcal{S}_{geo} \to 0$ forces $w_{ij} \to 0$ regardless of context similarity—geography dominates cluster structure. The components are:

$$\mathcal{S}_{geo} = \exp\left(-\frac{\mathrm{dist}(L_i, L_j)^2}{2\sigma_{geo}^2}\right) \tag{5}$$

$$\mathcal{S}_{temp} = \exp\left(-\frac{|T_i - T_j|}{\tau_{temp}}\right) \tag{6}$$

$$\mathcal{S}_{context} = \exp\left(-\frac{|F_i - F_j|}{\tau_F} - \frac{|E_i - E_j|}{\tau_E}\right) \tag{7}$$

$\mathrm{dist}(\cdot)$ is Haversine distance (meters); σ_{geo} is a characteristic radius approximating a boat's operating range (a few hundred meters to 1–2 km). The squared distance makes the Gaussian gate decay sharply, strongly penalizing distant links. The temporal kernel uses a first-order (non-squared) exponent because flood evolution has inertia. For context, similar reports ($\Delta F \approx 0, \Delta E \approx 0$) give $\mathcal{S}_{context} \to 1$, whereas dissimilar ones (one safe on a 3rd floor, one clinging to a rooftop) shrink it—correctly reflecting distinct rescue needs. Since $\exp(-a - b) = \exp(-a)\exp(-b)$, both conditions (similar flood *and* similar urgency) must hold jointly for a high $\mathcal{S}_{context}$. The looser $\tau_E = 0.35$ versus $\tau_F = 0.25$ makes the kernel more tolerant of a given urgency gap than of the same flood-depth gap, reflecting that the sentiment-derived E is noisier than the vision-derived F and therefore warrants a wider matching band; a sensitivity sweep (Sect. 5.3) confirms clustering is insensitive to both within a wide band, so this asymmetry is a mild modeling preference rather than a tuned parameter. The additive form's α is eliminated because $\mathcal{S}_{geo}$ is now a global factor.

Graph sparsification. Because $\mathcal{S}_{geo}$ decays quickly, most long-range edges carry negligible weight. We sparsify via either (i) a threshold θ (retain edges with $w_{ij} > \theta$) or (ii) a k-NN graph (each vertex keeps its k highest-weight neighbors). This reduces computation and removes spurious cross-zone links, since Modularity optimization degrades on dense, near-complete graphs.

4.3 Clustering via Louvain (Leiden Recommended)

We partition the vertex set into disjoint clusters $\{C_1, \ldots, C_k\}$ maximizing Modularity in the Reichardt–Bornholdt [18] resolution form:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \lambda \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{8}$$

where $A_{ij} = w_{ij}$, $k_i = \sum_j A_{ij}$ is the weighted degree, $m = \frac{1}{2} \sum_{i,j} A_{ij}$ is total graph weight, δ is the Kronecker delta, and $\frac{k_i k_j}{2m}$ is the null-model expected edge weight. The resolution λ controls granularity: $\lambda = 1$ is standard Modularity; $\lambda > 1$ splits more finely (a large flooded ward into distinct blocks); $\lambda < 1$ favors larger clusters.

Louvain is chosen because it (1) auto-discovers the cluster count—no a-priori K; (2) denoises structurally—fake reports lack strong edges and become singletons; (3) has observed near-$\mathcal{O}(N \log N)$ runtime for real-time use; and (4) is student-feasible via `python-louvain/networkx/igraph`. We recommend the **Leiden** [21] refinement, which guarantees well-connected communities, since a badly connected cluster would place the dispatch centroid away from the actual victims.

4.4 Cluster-Level Priority Scoring

Identifying clusters is insufficient: which cluster gets the first boat? We propose a priority function that fixes three mathematical flaws of a naive additive score: (a) scale bias, (b) vulnerability modeled as an additive rather than multiplicative term, and (c) early tanh saturation.

$$\mathcal{P}(C_k) = \mathcal{V}_{agg}(C_k) \cdot \Big(\omega_1 \widetilde{\mathcal{E}}_{agg}(C_k) + \omega_2 \widetilde{\mathcal{F}}_{max}(C_k) + \omega_3 \widetilde{\mathcal{N}}(C_k) \Big) \tag{9}$$

(a) Scale normalization. $\mathcal{E}_{agg}, \mathcal{F}_{max} \in [0,1]$ but $\mathcal{N}_{total} = \sum N_i$ is unbounded (possibly hundreds). Adding them directly lets population dominate. We normalize every component to $[0,1]$ before weighting $(\widetilde{(\cdot)})$. For the right-skewed population, we log-compress then min-max:

$$\widetilde{\mathcal{N}}(C_k) = \frac{\log(1 + \mathcal{N}_{total}(C_k))}{\log(1 + N_{\max})} \tag{10}$$

where $N_{\max}$ is a reference. A *dynamic* reference (largest current-window cluster) gives instantaneous relative ranking; a *fixed* reference is needed for across-time comparison. The dynamic mode is non-stationary and must be interpreted accordingly.

Risk-core components (all confidence-gated).

$$\mathcal{E}_{agg}(C_k) = \frac{1}{|C_k|} \sum_{v_i \in C_k} E_i \cdot C_i \tag{11}$$

$$\mathcal{F}_{max}(C_k) = \max_{v_i \in C_k} \left(F_i \cdot C_i\right) \tag{12}$$

$$\mathcal{N}_{total}(C_k) = \sum_{v_i \in C_k} N_i \cdot C_i \tag{13}$$

We use max (not mean) for flood depth by the "communicating-vessels" principle: the deepest point determines the cluster's survival risk; averaging would dilute the alarm. Crucially, C_i is multiplied *inside* the max so a low-confidence fake report claiming $F = 1.0$ cannot capture $\mathcal{F}_{max}$—restoring consistency with the confidence-gating of $\mathcal{E}$ and $\mathcal{N}$.

(b,c) Vulnerability amplification.

$$\mathcal{V}_{agg}(C_k) = 1 + \tanh\left(\frac{1}{s} \sum_{v_i \in C_k} V_i\right), \qquad \mathcal{V}_{agg} \in [1, 2) \tag{14}$$

Placing $\mathcal{V}_{agg}$ *outside* as a factor makes it a true amplifier: a cluster with no vulnerable individuals gets ≈ 1 (risk core unchanged), while one with many gets up to $2\times$. An additive term bounded in $[1, 2]$ would only add a near-constant offset—amplifying nothing. The scale factor s (e.g., $s = 10$) prevents early saturation: with raw $\tanh(\sum V_i)$, only 2–3 vulnerable individuals push tanh near 1, making a cluster with 1 indistinguishable from one with 50; dividing by s stretches the linear region while tanh still caps the value. An equivalent alternative is $1 + \log(1 + \sum V_i)$ with normalization.

Adjustable amplification cap. The fixed $2\times$ ceiling is a policy choice, not a mathematical necessity. To let command staff bound how far vulnerability may reorder the list—a genuine "many-healthy vs. few-vulnerable" trade-off—we generalize to

$$\mathcal{V}_{agg}(C_k) = 1 + (\mu - 1)\tanh\left(\frac{1}{s} \sum_{v_i \in C_k} V_i\right), \qquad \mu \in [1, 2], \tag{15}$$

so $\mu = 1$ disables amplification (pure risk core) and $\mu = 2$ recovers the default. All reported numbers use $\mu = 2$. A larger μ can push a small vulnerable-rich cluster above a large healthy one; exposing μ makes that ethical dial explicit rather than hard-coded.

Weights and range. $\omega_1, \omega_2, \omega_3$ (with $\sum \omega = 1$) are set by command staff via an Analytic Hierarchy Process (AHP) decision matrix [19] to switch tactical modes (population-first vs. depth-first). Since the core is normalized to $[0, 1]$ and $\sum \omega = 1$, the core lies in $[0, 1]$; multiplying by $\mathcal{V}_{agg} \in [1, 2)$ bounds $\mathcal{P}(C_k) \in [0, 2)$—convenient for ranking. Descending $\mathcal{P}(C_k)$ yields an actionable priority list; combined with cluster centroids it is an ideal input for downstream route optimization (A* cost-aware, multi-commodity routing).

On reusing F, E in two stages. F and E appear both in clustering (via $\mathcal{S}_{context}$) and prioritization (via $\mathcal{F}_{max}, \mathcal{E}_{agg}$). This is not erroneous double-counting: $\mathcal{S}_{context}$ measures pairwise *similarity* (whether two events share a situation), whereas $\mathcal{F}_{max}/\mathcal{E}_{agg}$ measure a cluster's *absolute severity* (for ranking). We acknowledge, however, that a cluster grouped by similar F will almost certainly have high $\mathcal{F}_{max}$, so $\mathcal{F}_{max}$ should be read as "the characteristic flood level of an already-homogeneous population," not a fully independent signal. We quantify this circularity in Sect. 5.7: removing $\mathcal{S}_{context}$ from the graph changes clustering (ARI drops) but barely moves the priority ranking, showing that $\mathcal{S}_{context}$ mainly aids *grouping* rather than covertly dictating the *ranking*.

5 Experiments

5.1 Setup

Dataset. Absent a real labeled ground-truth dataset for rescue clusters, we generate a *deterministic synthetic* dataset (seed = 42) modeled on Central Vietnam (Huế–Quảng Trị–Quảng Nam–Đà Nẵng; 15.7–17.1°N, 107.0–108.6°E) with **285 events** spanning **12 ground-truth labels**: 240 core events around **6 flood "islands"** ($\sim$40 each, labels 0–5), 20 scattered noise events (`gt`$= -1$, of which 5 are fake), and 25 narrative events—24 forming **6 stress-test groups** (labels 100–105) plus one isolated fake (`gt`$= -1$). The four scenarios each stress-test one design decision—S1: two rooftop-flood points $\sim$103 km apart (gating separation); S2: a cluster rich in vulnerable individuals ($\mathcal{V}_{agg}$ amplification); S3: an isolated fake report inflating 200 victims (C_i gate); S4: a large shallow-flood cluster vs. a small rooftop-flood cluster ($\mathcal{F}_{max}$). The narrative groups are deliberately *anchored at the island centres* and confined within each island's spatial footprint (offsets up to $\sim$0.83 km, comparable to the cores' own coordinate jitter—mean $\sim$0.32 km, max $\sim$0.85 km) yet carry distinct labels, which—as Sect. 5.2 details—caps the attainable ARI below 1.0 by construction.

Parameters. $\sigma_{geo} = 700\,\text{m}$; $\tau_{temp} = 45\,\text{min}$; $\tau_F = 0.25$; $\tau_E = 0.35$; $\beta = \gamma = 0.5$; edge threshold $\theta = 0.05$; k-NN $k = 12$; $\lambda = 1.0$; $s = 10$; $\omega = (0.34, 0.33, 0.33)$; confidence heuristic $(b_0, b_1, b_2) = (-0.2, 1.4, 0.9)$; corroboration window $(r_{\text{corrob}}, \Delta t_{\text{corrob}}) = (400\,\text{m}, 60\,\text{min})$. We distinguish *domain-set* parameters, fixed by operational meaning rather than tuned—σ_{geo} (boat range), τ_{temp} and Δt_{corrob} (flood-evolution timescale), $\lambda = 1$ (standard Modularity)—from *tunable* ones $(\tau_F, \tau_E, \beta, \gamma, \theta, k, s, \omega, b_*)$, whose robustness we probe in Experiments 2 and 5. Since a single synthetic dataset cannot support a train/validation/test split with statistical meaning, we report sensitivity sweeps instead of claiming optimality; learning these parameters on real data is future work.

Metrics. Adjusted Rand Index (ARI) [9] and Normalized Mutual Information (NMI) against ground truth; *geographic cluster diameter* (km, the largest intra-cluster point distance) for spatial coherence; and Modularity Q. The full pipeline is implemented in Python (`numpy`, `networkx`, `python-louvain`, `igraph`, `leidenalg`, `scikit-learn`).

Table 3. Gating vs. additive edge weighting (same dataset).

Form	ARI	NMI	Mean diam. (km)	Max diam. (km)	# Clusters
Additive	0.892	0.927	100.07	213.95	6
Multiplicative (gating)	**0.892**	**0.927**	**0.30**	**1.42**	27

5.2 Experiment 1 — Validating Six Design Decisions

(1A) Gating vs. additive. Both forms yield *identical* ARI = 0.892 and NMI = 0.927, but differ fundamentally in spatial coherence (Table 3, Fig. 1). The additive form produces clusters of ∼100 km mean diameter—useless for boat dispatch—while gating shrinks this to **0.30 km**. Two facts explain why the ARI is identical yet below 1.0. First, the ceiling below 1.0 is imposed *by the ground truth*, not by either weighting: the 24 narrative stress-test points (labels 100–105) are anchored at the six island centres and lie within each island's spatial footprint but carry distinct labels, so any spatially-driven method necessarily merges each into its host island. Measured alone, the 240 core events are recovered perfectly (ARI = 1.0) and the 24 narrative points are internally consistent (ARI = 1.0); it is only their co-location that pins the combined labeled-set ARI at 0.892. Second, the ARI is identical across the two forms because they induce the *same* partition on the 264 labeled events; they differ only on the 21 noise points (`gt`$= -1$), which gating isolates as spatial singletons while the additive form absorbs them into the large clusters (hence 6 vs. 27 total clusters). Since ARI/NMI mask out `gt`< 0, this difference is invisible to them. The key point stands: gating does *not* sacrifice labeled-cluster accuracy; it fixes the spatial geometry (100 km→0.30 km) *and* isolates noise, neither of which ARI can reward.

(1B) Scale normalization. Without normalization, the top-ranked cluster is simply the most populous (216 people, raw core 71.65)—population dominates. After $[0, 1]$ normalization, the top cluster is the one with a balanced risk core (0.82), scoring $\mathcal{P} = 1.52$.

(1C) Multiplicative vs. additive vulnerability. For scenario cluster S2 (vulnerable-rich, $\mathcal{V}_{agg} = 1.97$), the additive form gives $\mathcal{P}_{add} = 1.66$ and the multiplicative form $\mathcal{P}_{mult} = 1.36$. More important is the *differential behavior*: for clusters with no vulnerable individuals ($\mathcal{V}_{agg} = 1$) both forms coincide exactly, but as $\mathcal{V}_{agg}$ grows the multiplicative form scales with the risk core (true amplification) whereas the additive form only adds a near-constant offset.

(1D) tanh *anti-saturation.* Table 4 shows raw $\tanh(\sum V_i)$ saturates (≈ 2.0) by $\sum V_i = 3$, losing all discrimination, whereas $\tanh(\sum V_i/10)$ keeps growing to $\sum V_i = 50$ (Fig. 2).

(1E) Confidence gate on population. In scenario S3 (a fake report inflating 200 victims, $C_i = 0.45$), the cluster's ungated population is 200; after multiplying by

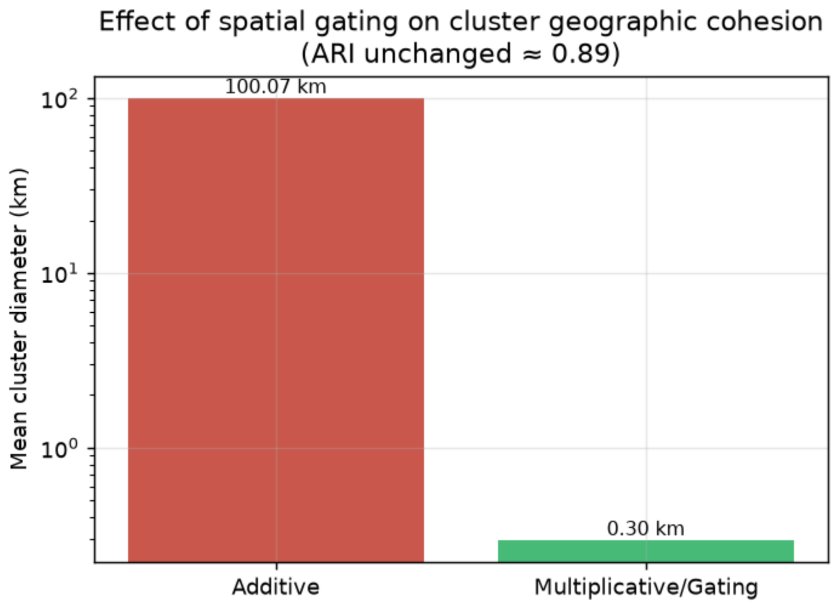

Fig. 1. Additive weighting merges spatially distant events into sprawling clusters; the gating form yields spatially compact operational zones at equal ARI.

Table 4. Vulnerability factor $\mathcal{V}_{agg}$ with and without scale $s = 10$.

$\sum V_i$	1	3	10	30	50
$1 + \tanh(\sum V_i)$	1.76	2.00	2.00	2.00	2.00
$1 + \tanh(\sum V_i/10)$	1.10	1.29	1.76	2.00	2.00

C_i it drops to **90—a 55% reduction** (Fig. 3), preventing a fabricated report from topping the priority list.

(1F) Confidence gate on maximum flood depth. The same S3 report claims $F = 0.99$ with $C_i = 0.45$. Under a plain $\max F_i$ it would seize $\mathcal{F}_{max} = 0.99$; under $\max(F_i \cdot C_i)$ it drops to **0.45**, restoring consistency: all three risk-core components $(\mathcal{E}, \mathcal{F}, \mathcal{N})$ are confidence-gated, closing the last loophole for a single untrusted report to manipulate the ranking.

5.3 Experiment 2 — Sensitivity Analysis

Resolution λ: ARI is stable at 0.892 for $\lambda \leq 1.5$; it degrades at $\lambda = 2.0$ (ARI 0.83) and $\lambda = 3.0$ (ARI 0.67) as clusters fragment. Recommended safe range: $\lambda \in [0.5, 1.5]$ (Fig. 4, right). Radius σ_{geo} directly trades diameter for cluster count: ARI holds at 0.892 across a wide range while mean diameter grows from

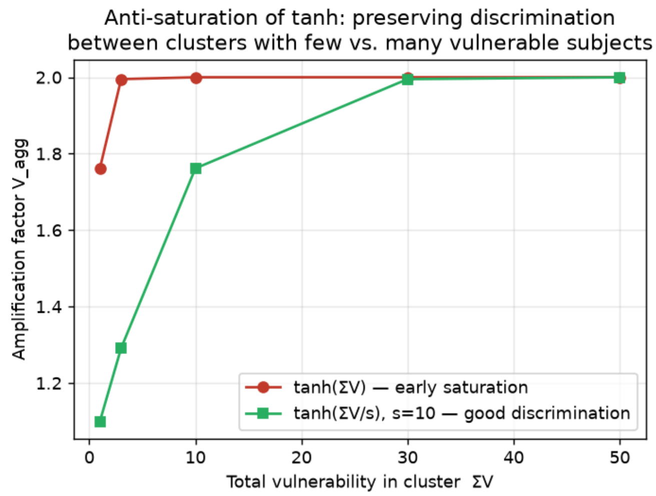

Fig. 2. Without scaling, $\tanh(\sum V_i)$ saturates by $\sum V_i \approx 3$; the $s = 10$ variant preserves discrimination up to $\sum V_i \approx 50$.

0.28 km ($\sigma_{geo} = 200$ m) to 1.59 km ($\sigma_{geo} = 4000$ m) (Fig. 4, left); σ_{geo} should match the real operating range of the rescue unit. Scale s: the spread of $\mathcal{V}_{agg}$ narrows as s grows ($s = 1$ spans $1.0 \to 2.0$; $s = 20$ only $1.0 \to 1.78$); $s = 10$ gives good discrimination up to $\sum V_i \approx 50$.

To answer the "too many free parameters" concern, we extend the sweep to the context and balance parameters. The context sensitivities τ_F, τ_E are *inert* for clustering accuracy: ARI stays at 0.892 across the full grid $\tau_F, \tau_E \in [0.15, 0.5]$, because the geographic gate dominates cluster structure regardless of how sharply context similarity decays. The balance β/γ matters only near the extremes: ARI holds at 0.892 for $\beta \leq 0.5$ (context-inclusive) but drops to 0.7855 once $\beta \geq 0.9$ starves the context term—consistent with the ablation of Sect. 5.7. This shows most parameters are either domain-set (Table's "domain" group) or provably inert on this dataset, leaving only $\lambda, \sigma_{geo}, \beta/\gamma$ as accuracy-relevant tuning knobs, all with safe operating ranges reported above.

5.4 Experiment 3 — Louvain vs. Leiden

Across **10 seeds**, both Louvain and Leiden yield **zero badly-connected communities**, at ARI 0.892 and Modularity 0.8311. This is an honest, noteworthy finding: the gating mechanism itself produces spatially coherent subgraphs, preempting the broken-community risk, so Louvain already suffices here. Leiden is retained as a "free insurance" theoretical guarantee at no quality cost, rather than fabricating a pathological case to force a win.

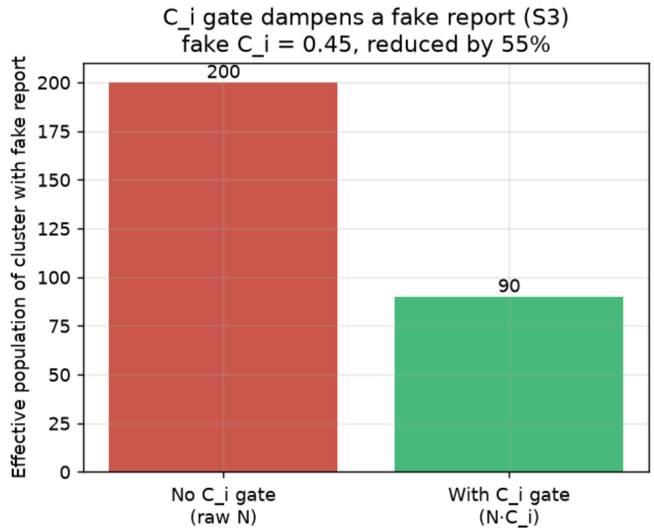

Fig. 3. Confidence gating: a fake report ($C_i = 0.45$) inflating 200 victims is reduced to an effective 90, and its claimed $F = 0.99$ to 0.45.

5.5 Experiment 4 — Baseline Comparison

Table 5 compares against three *fair* baselines run on the *same* gating graph (Spectral clustering [23], HDBSCAN [3], Agglomerative) alongside purely geometric baselines (K-Means, DBSCAN on raw coordinates); see Fig. 5.

On the same affinity/distance matrix, Louvain/Leiden remain superior: (i) **Spectral** attains only ARI 0.339—the sparse gating graph hinders spectral partitioning; (ii) **HDBSCAN** reaches ARI 0.890 but finds fewer clusters (11 vs. 27) at 25 km mean diameter—inflated by a catch-all noise bin (∼36 labeled points) plus one scattered residual cluster, as it *fragments* two islands rather than forming coherent zones; (iii) **Agglomerative** matches Louvain on ARI/NMI but needs K in advance—infeasible in a disaster. Since HDBSCAN uses the same graph yet underperforms, the advantage stems from the *combination* of the gating graph and Modularity optimization (auto-K, spatial coherence), not from either component alone.

5.6 Experiment 5 — Ranking Stability (Kendall's τ)

A potential critique: command staff set ω manually; if the $\mathcal{P}(C_k)$ ranking is too sensitive to ω, the priority list becomes arbitrary. We perturb ω around the default $(0.34, 0.33, 0.33)$, renormalize to $\sum \omega = 1$, and measure Kendall's τ [12] against the baseline ranking (200 Monte-Carlo trials per level; Table 6, Fig. 6).

At realistic perturbation (±0.05–±0.10), τ stays above 0.93 and the top-3 clusters are nearly unchanged (99–100%). Even at an extreme ±0.20 (altering

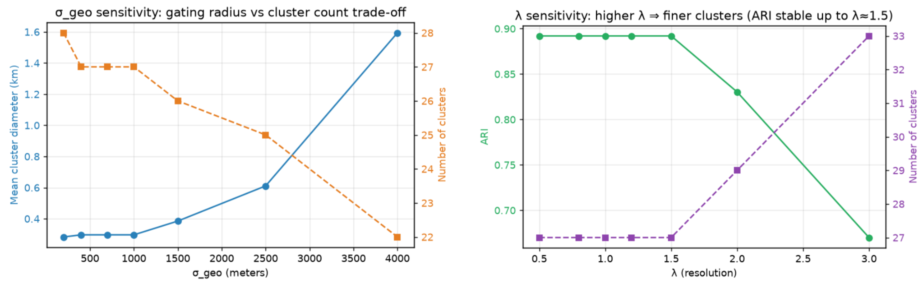

Fig. 4. Sensitivity sweeps. Left: σ_{geo} trades cluster diameter against count at stable ARI. Right: resolution λ is safe within $[0.5, 1.5]$.

Table 5. Baseline comparison. "Same graph?" = run on the gating affinity/distance matrix.

Method	# Clus.	ARI	NMI	Mean diam. (km)	Same graph?	Needs K?
Louvain (gating)	27	**0.892**	**0.927**	**0.30**	✓	No
Leiden (gating)	27	**0.892**	**0.927**	**0.30**	✓	No
Spectral (gating, K=27)	27	0.339	0.727	14.11	✓	Yes
HDBSCAN (dist $1-w$)	11	0.890	0.922	25.08	✓	No
Agglomerative (dist $1-w$)	27	0.892	0.927	0.30	✓	Yes
K-Means (K=12, coords)	12	0.688	0.834	49.21		Yes
K-Means (K=3, coords)	3	0.433	0.630	102.04		Yes
DBSCAN (eps 0.3, coords)	15	0.644	0.783	15.12		No
DBSCAN (eps 0.6, coords)	7	0.730	0.873	32.27		No

nearly 60% of ω), mean τ is 0.957, showing that $\mathcal{P}(C_k)$ produces a stable ranking. Stability also holds against the *clustering* step, not just the weights: sweeping σ_{geo} over $[400, 1200]$ m leaves the priority ranking essentially fixed (Kendall's $\tau = 1.0$ up to 900 m, 0.9815 at 1200 m where two clusters merge), so the ranking does not hinge on a single graph-construction choice.

5.7 Experiment 6 — Context Circularity Ablation

A reviewer may worry that reusing F, E in both $\mathcal{S}_{context}$ (clustering) and $\mathcal{F}_{max}, \mathcal{E}_{agg}$ (ranking) lets the context term covertly dictate the final priority list. We test this by removing $\mathcal{S}_{context}$ from the edge weight (setting γ=0, geographic–temporal gating only) and comparing both the clustering and the induced ranking against the full model. Removing context *does* change clustering—ARI drops from 0.892 to 0.7855, NMI from 0.927 to 0.8741, and the partition fragments from 27 to 30 clusters—confirming $\mathcal{S}_{context}$ genuinely aids grouping. Yet the priority ranking is barely affected: over the 27 clusters matched by centroid, Kendall's τ between the full-model and context-ablated rankings is 0.9829, and all top-5 clusters are preserved. This dissociation shows $\mathcal{S}_{context}$ mainly improves *who is grouped with whom*, not *which cluster is ranked first*; the ranking is driven by

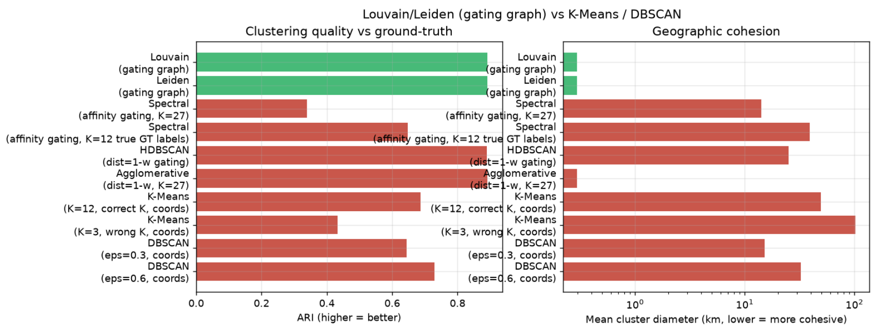

Fig. 5. Baseline comparison on ARI and geographic cluster diameter. Louvain/Leiden on the gating graph are Pareto-optimal: high ARI at minimal diameter, without needing K.

Table 6. Ranking stability under ω perturbation (200 trials/level).

Perturbation	Mean τ	Min τ	Top-3 preserved (%)
± 0.05	**0.994**	0.977	**100.0**
± 0.10	0.986	0.937	99.0
± 0.20	0.957	0.841	76.5

the absolute-severity terms $\mathcal{F}_{max}, \mathcal{E}_{agg}$ as intended, so the reuse is not vicious double-counting.

5.8 Experiment 7 — Does Vulnerability Weighting Actually Help the Vulnerable?

The deeper question behind Gap 2 (the vulnerability blind spot) is not whether the vulnerability index V_i changes the ranking (it does, by construction) but whether it produces a *better rescue outcome* for vulnerable populations. We simulate a discrete-event dispatch: 3 boats at 30 km/h with a 15 min service time per cluster, serving clusters in priority order, and measure the time until vulnerable victims (those with high V_i) are reached. We compare three priority policies: the full multiplicative $\mathcal{P} = \mathcal{V}_{agg} \cdot (\dots)$, an additive variant $\mathcal{P} = \mathcal{V}_{agg} + (\dots)$, and a vulnerability-blind policy (V_i removed). Removing V_i delays the mean time-to-vulnerable from 146.5 to 163.6 min—a **10.4% improvement** attributable to vulnerability weighting—and lowers harm-weighted response time by 8.7%. The multiplicative form buys this equity gain at a modest cost in mean all-victim arrival (942.9 vs. 719.6 min for the blind policy), an explicit and defensible fairness–efficiency trade-off rather than a mere reordering. The additive variant reaches the vulnerable slightly faster (133.3 min) but weakens the coupling between vulnerability and severity that the multiplicative gate is designed to preserve; we report both so the trade-off is transparent.

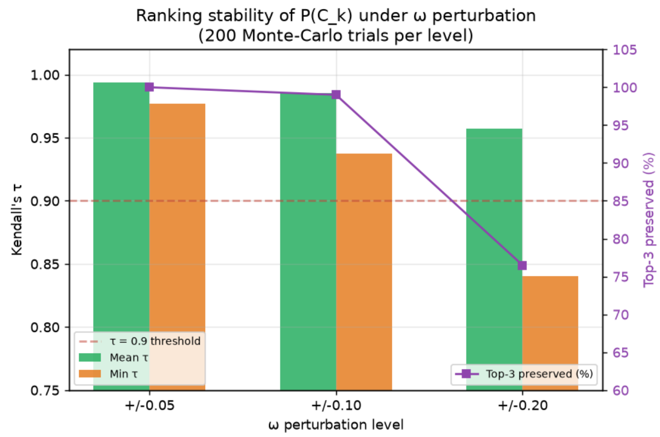

Fig. 6. Ranking stability: Kendall's τ stays above 0.93 at realistic perturbation (±0.05–±0.10), with the top-3 clusters essentially unchanged.

5.9 Experiment 8 — Confidence Detector and Adversarial Robustness

We evaluate the confidence heuristic C_i as a fake-report detector and probe its worst case. Treating low C_i as a fake-flag over all 285 events, C_i separates the injected fakes from real reports with **ROC-AUC** 0.9651 (mean C_i 0.50 for fakes vs. 0.92 for real). This is a screening signal, not a guarantee, and we deliberately stress its failure mode. A naive fake (isolated, no image) scores $C_i = 0.45$ and is easily down-weighted; but an adversary who *adds a forged image* raises it to 0.77, who *stages coordinated corroboration* raises it to 0.74, and who does *both* reaches $C_i = 0.92$—indistinguishable from the real-report mean. Because "independence" of corroboration is approximated purely by spatiotemporal proximity (no persistent-account infrastructure), the heuristic is robust to lone spammers but *not* to a resourced coordinated adversary. We therefore present C_i as a first-line filter that must be paired with account-level trust or cross-channel verification before it can be relied on adversarially, and we state this limitation explicitly rather than overclaiming detection.

5.10 Experiment 9 — Decomposing Cluster Quality: Completeness Beyond ARI

ARI saturates near the top of our leaderboard (Louvain, Leiden, Agglomerative, and even HDBSCAN all cluster around 0.89–0.892), which risks hiding real quality differences. We therefore *decompose* agreement into the homogeneity/completeness pair (whose harmonic mean, the V-measure, equals the arithmetic-averaged NMI already tabulated in Tables 3 and 5—so the added discrimination comes from the two *components*, not from a new aggregate metric).

The decomposition is far more discriminative on the failure that matters here—*completeness*, i.e. whether a ground-truth event is split across clusters. It exposes two failures ARI blurs. First, at the low end, Spectral clustering fragments events: completeness 0.595 (V-measure 0.727) against Louvain's 1.0 (V-measure 0.927). Second, and more tellingly, among the four leaders that ARI collapses into a near-tie (Louvain, Leiden, Agglomerative, and HDBSCAN, all within 0.002 ARI), completeness still separates them: Louvain/Leiden/Agglomerative reach a perfect 1.0 while HDBSCAN drops to 0.929 because it *fragments* ground-truth events—dumping ∼36 labeled points into its noise bin and splitting one island across residual clusters—a 0.07 completeness gap where ARI shows barely 0.002. (Its homogeneity, 0.915, is if anything *higher* than Louvain's 0.864: HDBSCAN's clusters stay pure precisely because it discards borderline points rather than merging classes, so the failure surfaces in completeness, not homogeneity.) (The full-range ARI spread of 0.55 is real but sits entirely at the low end—Spectral 0.339, K-Means 0.688—so it cannot rank the leaders.) We therefore report the homogeneity/completeness decomposition alongside ARI throughout.

6 Discussion

Cross-disciplinary impact. The framework delivers synergistic value across telecommunications/edge (resilience under network collapse; a compact JSON event descriptor—id, coordinates, an epoch timestamp, and the L, T, F, E, N, V, C fields—measures a deterministic *100–111 bytes* across all 285 events (min/median/max, whitespace-stripped), versus the raw multimedia post it summarizes, a MB→sub-KB reduction that keeps the mesh/LoRa uplink viable), data science/AI (recasting static classification as network-topology mining with quantified propagation risk), and rescue ethics (integrating a vulnerability index into the priority function, reshaping equity). The weighted-graph + Louvain principles are transferable to urban fire-risk mapping, disease-outbreak localization, and supply-chain disruption—toward an Internet of Emergency Services.

Scope relative to the project proposal. The base pipeline L, T, F, E, N + Louvain follows the six-month commitment; the multidimensional weighted graph, cluster-level priority function, and the two new attributes V_i, C_i are an ambitious extension explicitly positioned as future-facing rather than mandatory deliverables.

6.1 Threats to Validity

Internal: results use self-generated synthetic data with designed, well-separated flood islands, so the ARI level partly reflects data separability, not purely method strength; note the reported 0.892 is itself a by-construction ceiling (co-located narrative labels, Sect. 5.2) rather than a residual clustering error. **External:** only one region (Central Vietnam) and one hazard (flood) are tested; σ_{geo}, τ need recalibration elsewhere. **Construct:** ARI/NMI measure structural agreement, *not* rescue-decision quality; we partly address this with the outcome-oriented queue simulation of Sect. 5.8 (mean time-to-vulnerable-victim), though it too runs on synthetic demand. Because ARI saturates near the top methods (Sect. 5.10), we also report homogeneity/completeness to keep the comparison discriminative. **Conclusion:** Experiment 3 uses 10 seeds, but other experiments mostly use seed = 42; reporting mean ± std across seeds would strengthen every headline number.

7 Limitations and Future Work

Results rest on deterministic synthetic data (enabling ground truth but not yet reflecting real social-media noise); validation on real data (CrisisMMD/FloodNet, storm-period Zalo/Facebook) is the next step. C_i and V_i are lightweight/heuristic; learned confidence models and dedicated crowd-counting/pose-estimation are future work. Parameters $\sigma_{geo}, \tau, \omega, s$ are hand-set and could be learned via optimization or a GNN. Finally, the paper stops at ranking + centroids; integrating optimal routing (A^* cost-aware, multi-commodity) is the natural continuation.

8 Conclusion

We proposed an end-to-end framework combining Edge AI and weighted graph theory for flood-rescue event clustering and prioritization, addressing three gaps: missing physical attributes in edge weights, the vulnerability blind spot, and the absence of cluster-level priority. Two key methodological contributions—a *multiplicative/gating* weight function so geography dominates cluster structure, and a priority function with an *equity amplifier factor*—are validated quantitatively: gating cuts cluster diameter from 100 km to 0.30 km at unchanged ARI 0.892; the confidence gate blocks 55% of phantom population from fake reports; on the same gating graph Louvain beats Spectral (0.339) and HDBSCAN (0.890 at 25 km diameter) without needing K; and Monte-Carlo tests confirm a stable ranking (Kendall's τ mean 0.99, min 0.94 at ±0.10; top-3 preserved 99%). The framework offers a rigorous, equitable, and operationally feasible foundation for next-generation intelligent disaster-response platforms that stay functional even when telecommunication infrastructure degrades.

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